



Data Science - Project Course

Lecture 4: Linear Regression Analysis

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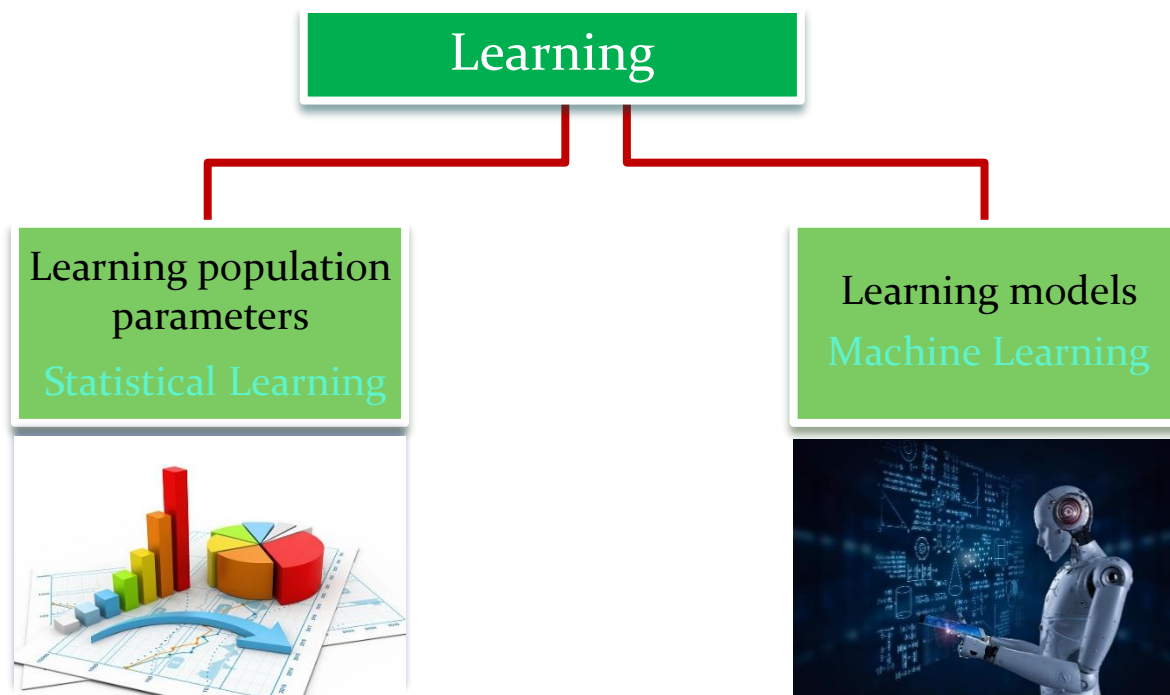
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Learning Strategies



There are two types of learning concepts:



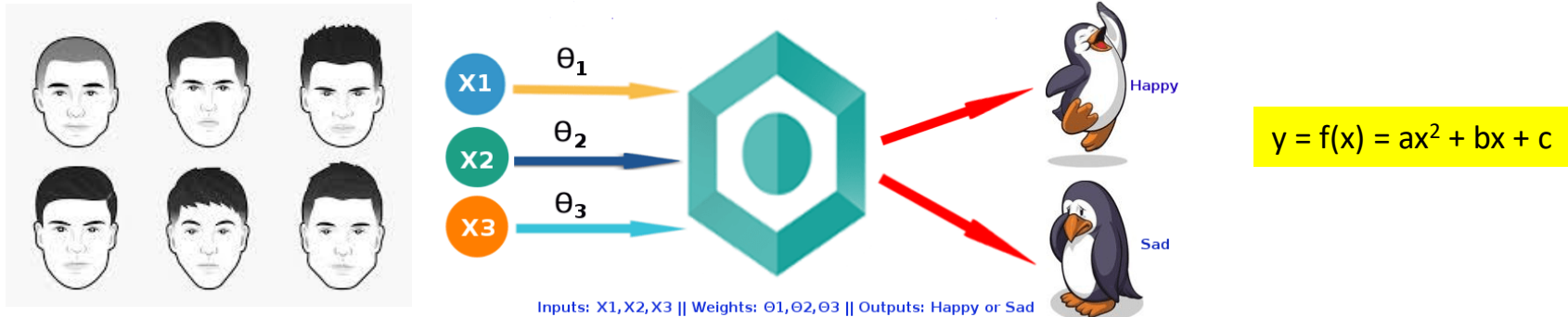
Statistical Learning



Usually assumes certain properties of the population from which we draw samples:

-  Observation come from a normal population.
-  Sample size is small.
-  Population parameters like mean, variance, etc. are hold good.
-  Requires measurement equivalent to interval scaled data.

Machine Learning



- Does not under any assumption
- Works well with high volume high dimensional data

Important Point

This learning strategy needs a very large sample data



Relationship Analysis



Relationship Analysis

- **Example: Wage Data**

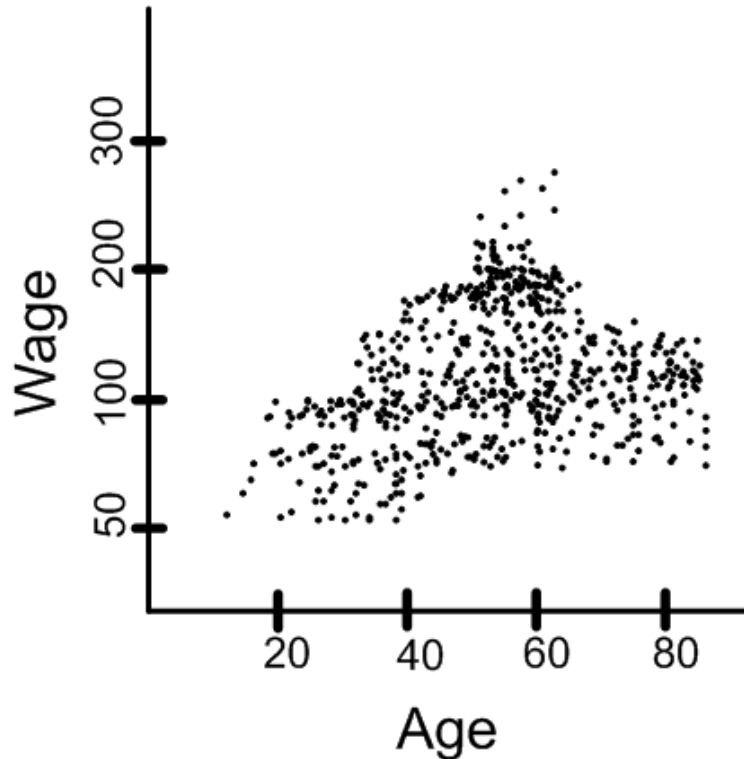
A large data regarding the wages for a group of employees from the eastern region of India is given.

In particular, we wish to understand the following relationships:

- *Employee's age and wage:* How wages vary with ages?
- *Calendar year and wage:* How wages vary with time?
- *Employee's age and education:* Whether wages are anyway related with employees' education levels?

Relationship Analysis

- Example: Wage Data
 - Case I. Wage versus Age
 - From the data set, we have a graphical representations, which is as follows:



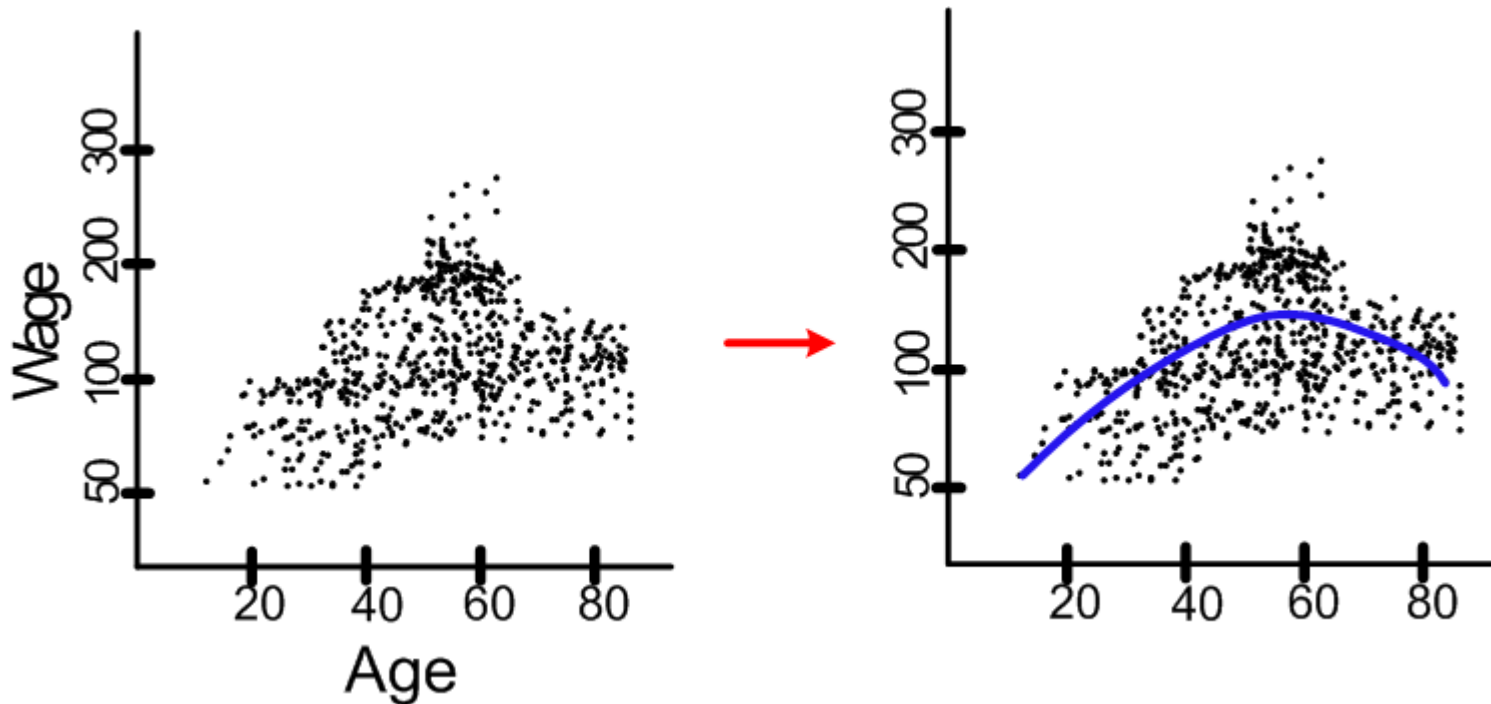
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How wages vary with ages?

Relationship Analysis



- Example: Wage Data
 - *Employee's age and wage: How wages vary with ages?*



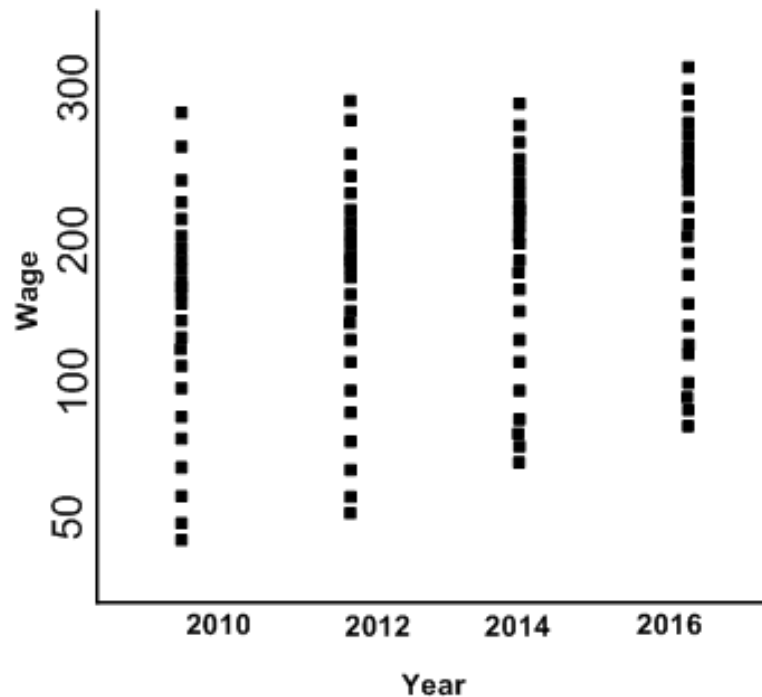
Interpretation: On the average, wage increases with age until about 60 years of age, at which point it begins to decline.

Relationship Analysis

- Example: Wage Data

- Case II. Wage versus Year

- From the data set, we have a graphical representations, which is as follows:



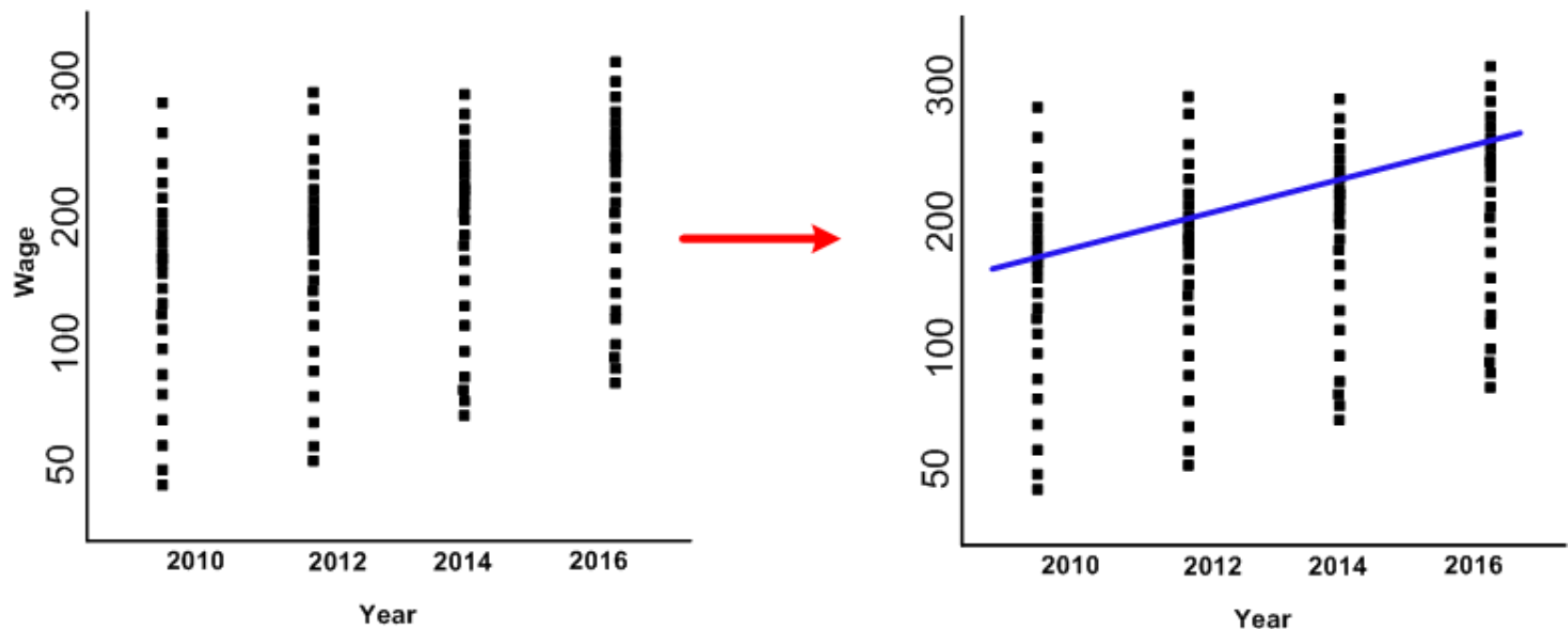
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How wages vary with time?

Relationship Analysis



- Example: Wage Data
 - *Wage and calendar year: How wages vary with years?*

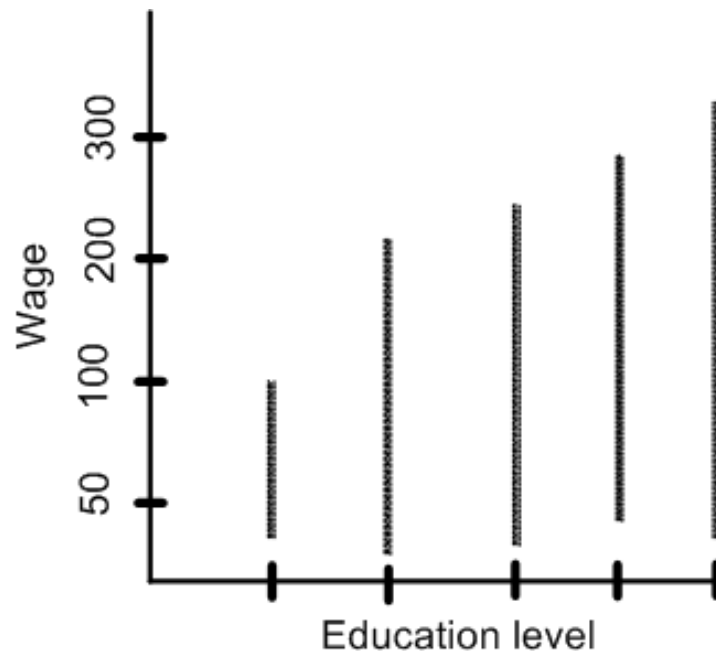


Interpretation: There is a slow but steady increase in the average wage between 2010 and 2016.

Relationship Analysis



- Example: Wage Data
 - Case III. Wage versus Education
 - From the data set, we have a graphical representations, which is as follows:

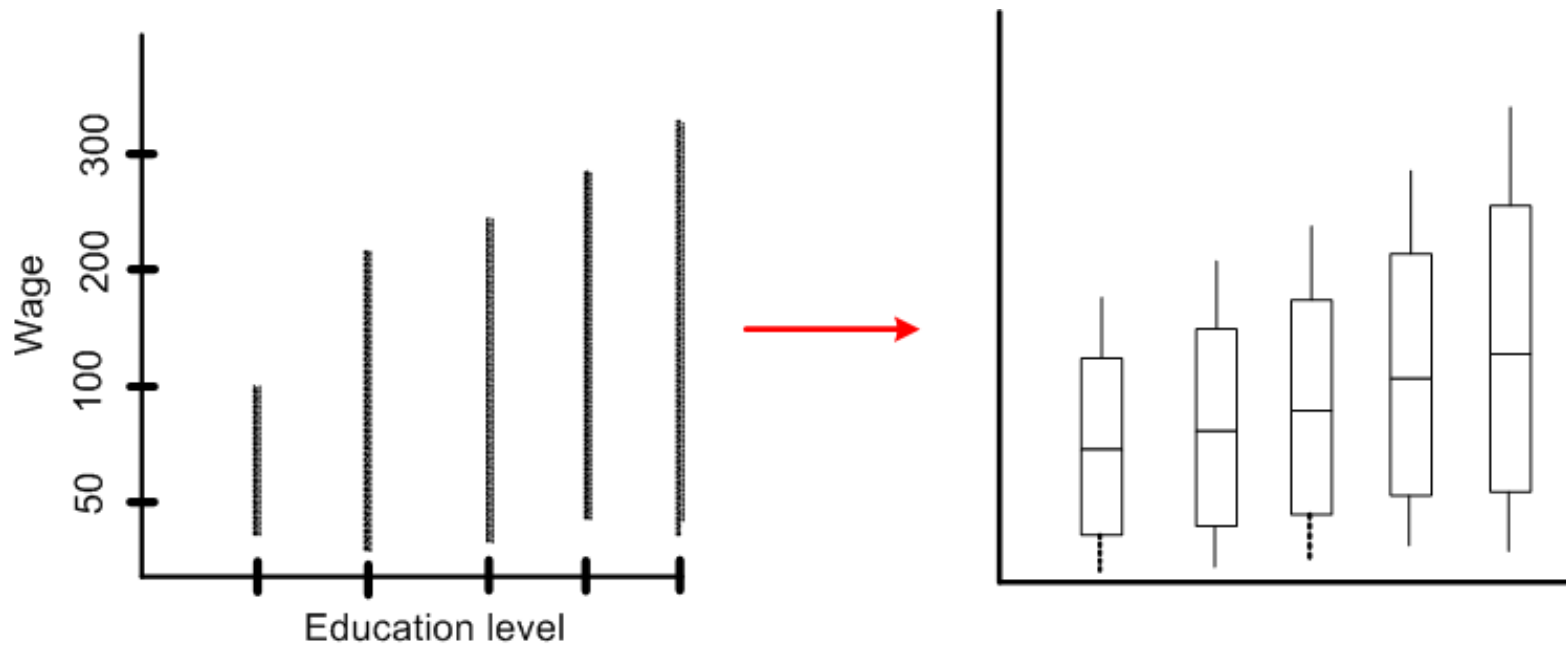


?

Whether wages are related with education?

Relationship Analysis

- Example: Wage Data
 - *Wage and education level: Whether wages vary with employees' education levels?*



Interpretation: On the average, wage increases with the level of education.

Relationship Analysis



What more information can we get?

Whether wage has
any association
with both year and
education level?

Given an employee's
wage can we predict
his age?

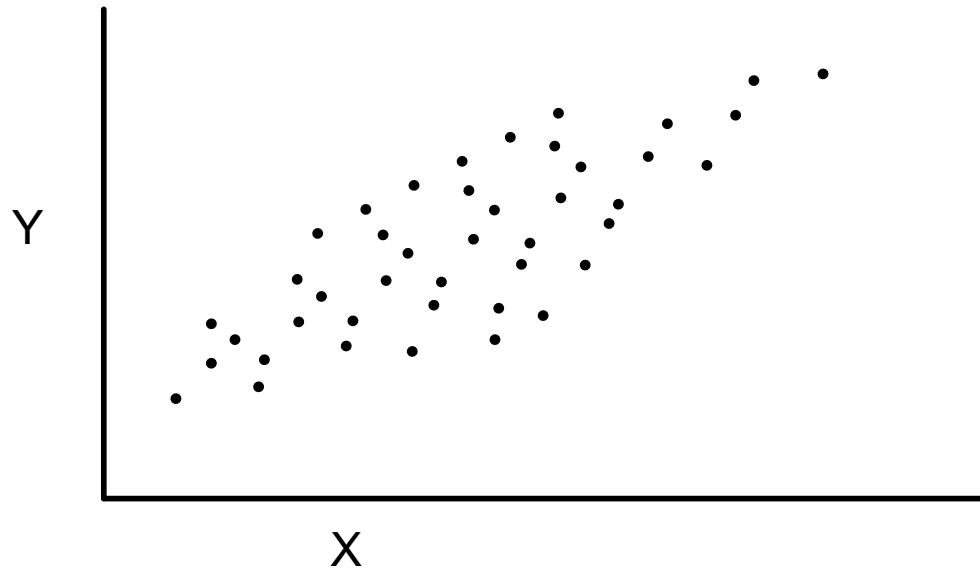


... and what's more?



Regression Analysis to find Relationships

A curious Question!

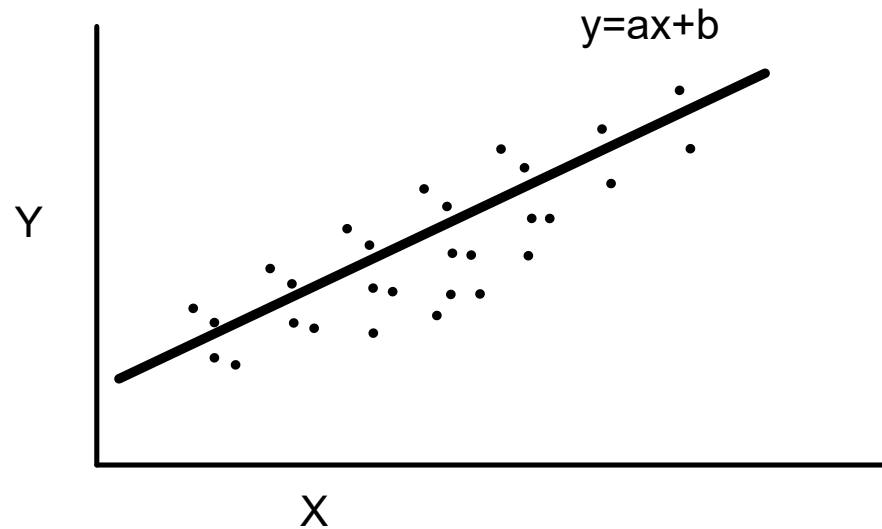


Suppose there are countably infinite points in the XY plane. We need a huge memory to store all such points.

Is there any way out to store this information with a least amount of memory?

Say, with two values only.

Yahoo!



Just decide the values of **a** and **b**
(as if storing one point's data only!)

Note: Here, tricks was to find a relationship among all the points.

Measures of Relationship



Univariate Population

<i>Temperature</i>	20	30	21	18	23	45	52
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Bivariate Population

<i>Temperature</i>	20	30	21	18	23	45	52
<i>Pressure</i>	1	1.5	1.05	0.96	1.2	2.5	2.8

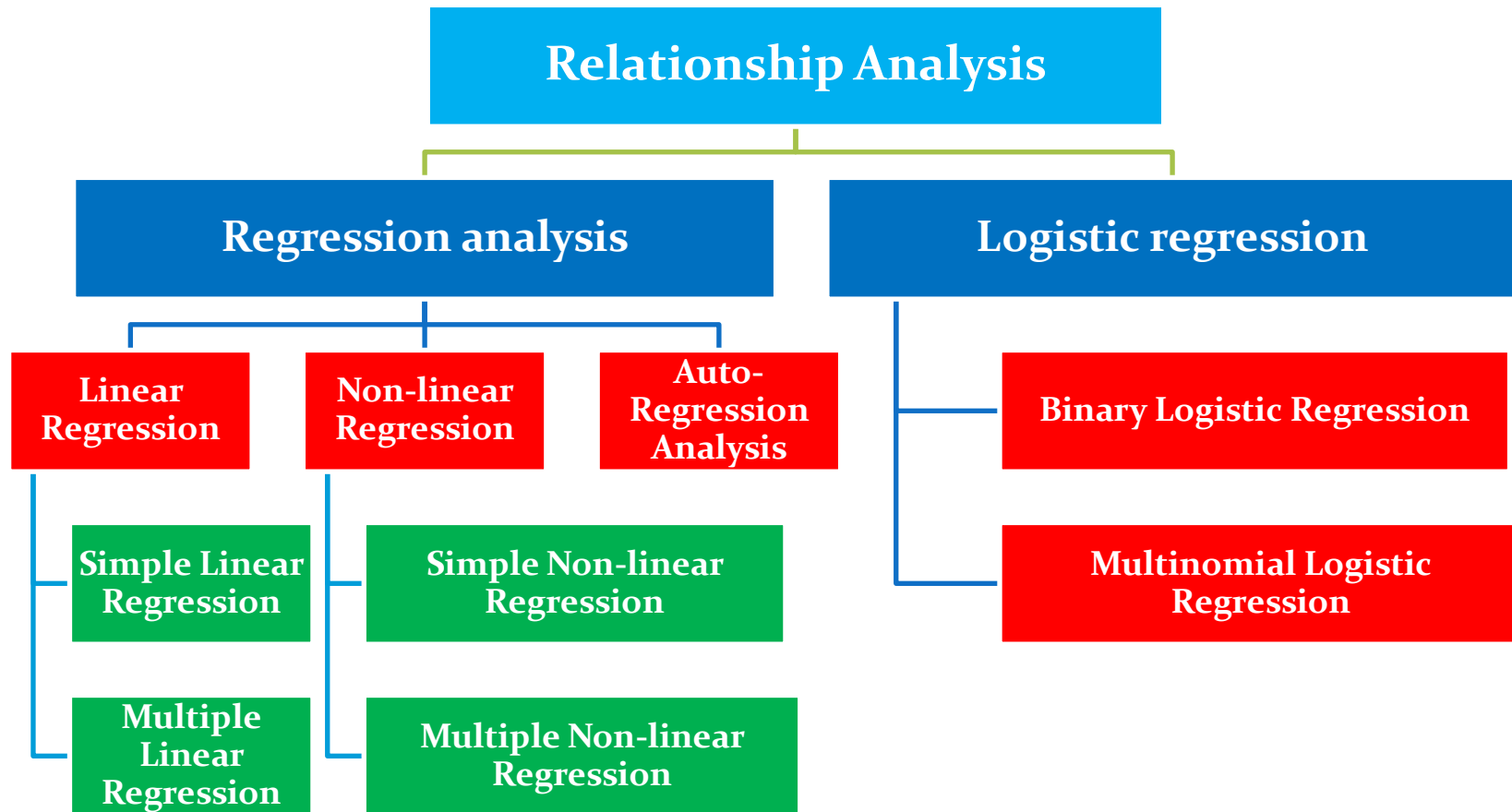


Multivariate Population

<i>Temperature</i>	20	30	21	18	23	45	52
<i>Pressure</i>	1	1.5	1.05	0.96	1.2	2.5	2.8
<i>Volume</i>	20	30	21	18	23	45	52



Measures of Relationship





Regression Analysis

Definition

The regression analysis is a statistical method to deal with the formulation of mathematical model depicting relationship amongst variables, which can be used for the purpose of prediction of the values of **dependent variable**, given the values of **independent variable(s)**.

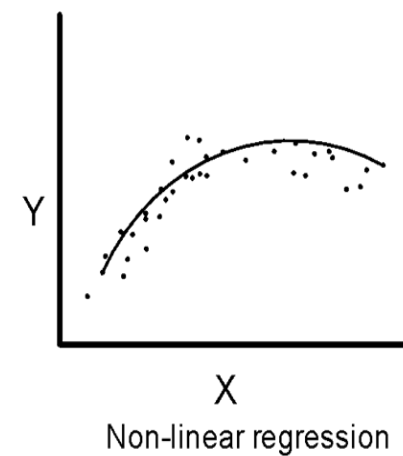
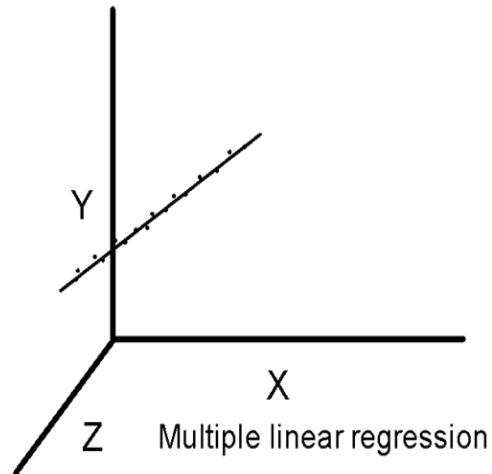
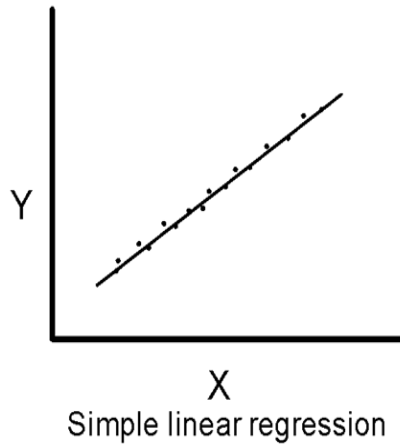
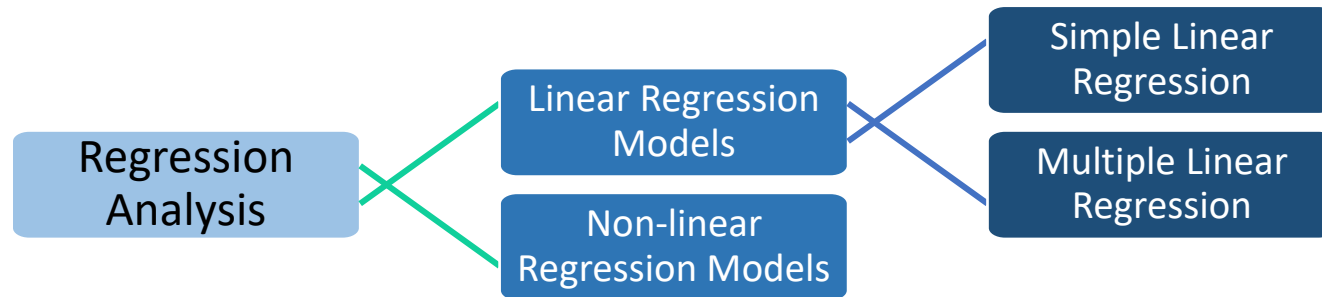
A Simple Example



How Exam Score is related to Hours of Study?

<i>Hours Study</i>	<i>Exam Score</i>
3	80
5	90
2	75
6	80
7	90
1	50
2	65
7	85
1	40
7	100

Regression Analysis





Simple Linear Regression

Simple Linear Regression Model

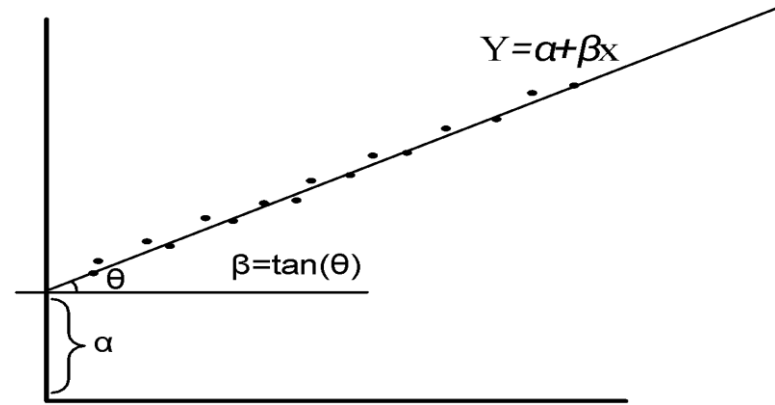
In simple linear regression, we have only two variables:

Dependent variable:

Also called *Response*, usually denoted as Y

Independent variable:

Also called *Regressor*, usually denoted as x

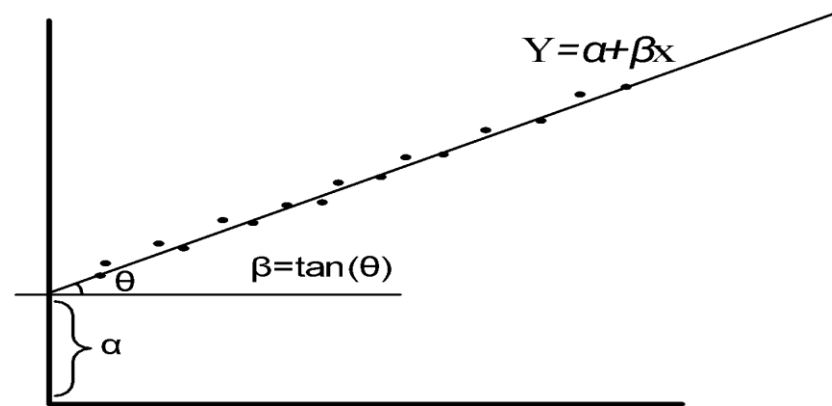


Linear regression

A reasonable form of a relationship between the Response Y and the Regressor x is the linear relationship, that is in the form $Y = \alpha + \beta x$

Simple Linear Regression Model

In simple linear regression, we have only two variables:



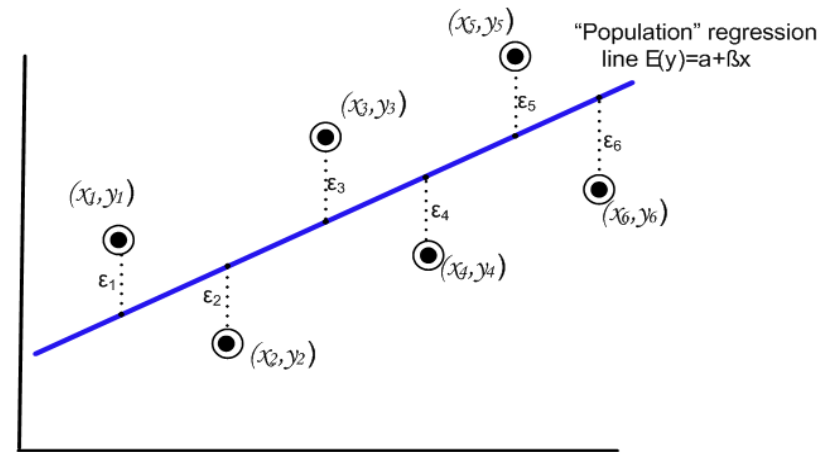
Note

- There are infinite number of lines (and hence α_s and β_s)
- The concept of regression analysis deal with finding the best relationship between Y and x (and hence best fitted values of α and β) quantifying the strength of that relationship.

Regression Analysis

Given: The set $[(x_i, y_i), i = 1, 2, 3, \dots, n]$ of data involving n pairs of (x, y) values

Objective: To find “true” or population regression line, such that $Y = \alpha + \beta x + \epsilon$



Here, ϵ is a random variable with $E(\epsilon) = 0$ and $var(\epsilon) = \sigma^2$. The quantity σ^2 is often called the **error variance**.

Note:

$E(\epsilon) = 0$ implies that at a specific x , the y values are distributed around the “true” regression line $Y = \alpha + \beta x$ (i.e., the positive and negative errors around the true line is reasonable).

α and β are called **regression coefficients**.

α and β values are to be estimated from the data.

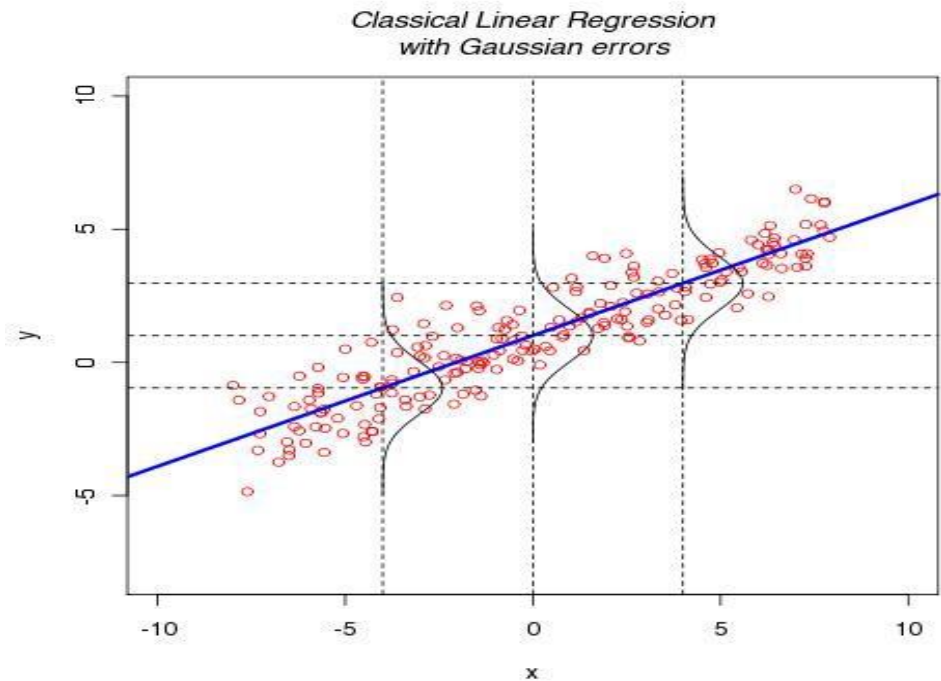
Assumption on the model

- The Linear Regression Model for i th observation

$$Y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, 2, \dots, n.$$

- Assumptions:

- $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2$
- $Cov(\epsilon_i, \epsilon_j) = 0, i \neq j$
- ϵ_i (iid) $\sim N(0, \sigma^2)$



Assumption on the model

Consequent assumptions on Y:

- $E(Y_i) = \alpha + \beta x_i$
- $Var(Y_i) = \sigma^2$
- Y_i 's are independent and normally distributed.

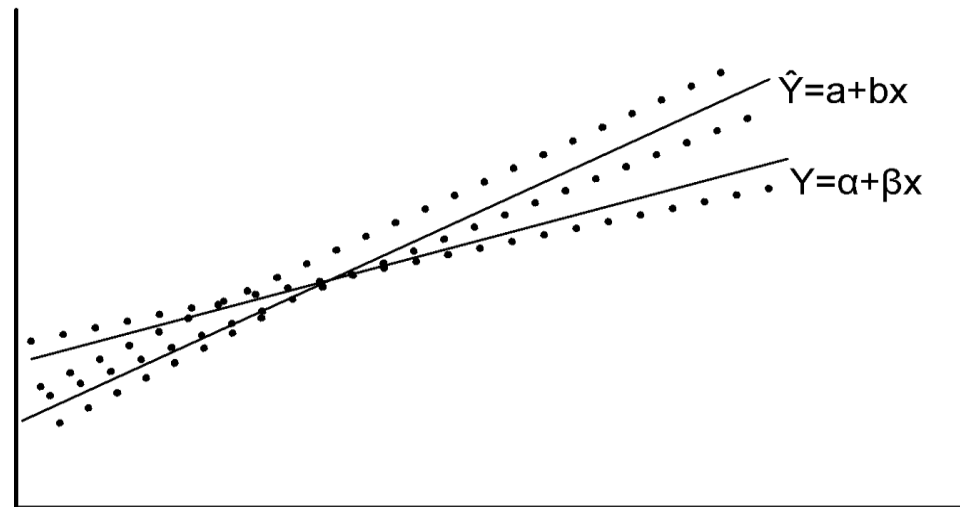
True versus Fitted Regression Line

📍 The task in regression analysis is to estimate the regression coefficients α and β .

📍 Suppose, we denote the estimates a for α and b for β . Then the fitted regression line is

$$\hat{Y} = a + bx$$

where, $\hat{Y}$ is the predicted or fitted value

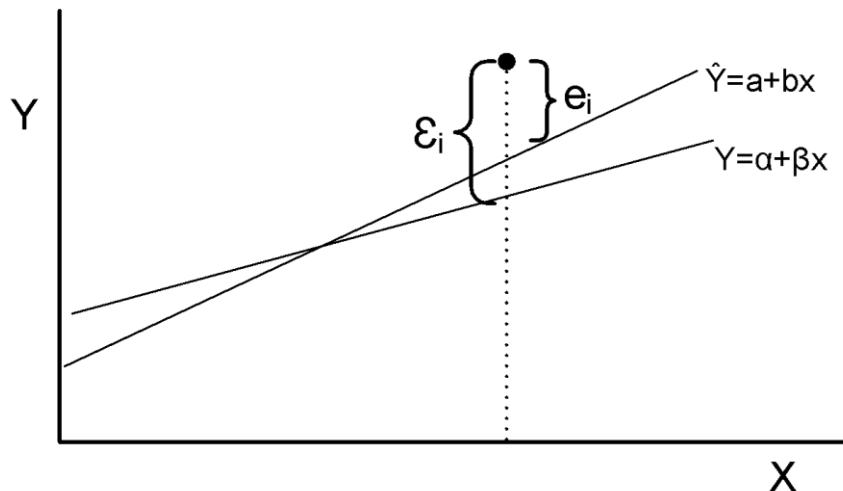


Least Square Method to estimate α and β

Concept of Residuals

This method uses the concept of residual. A residual is essentially an error in the fit of the model $\hat{Y} = a + bx$. Thus, i^{th} residual is

$$e_i = Y_i - \hat{Y}_i, i = 1, 2, 3, \dots, n$$



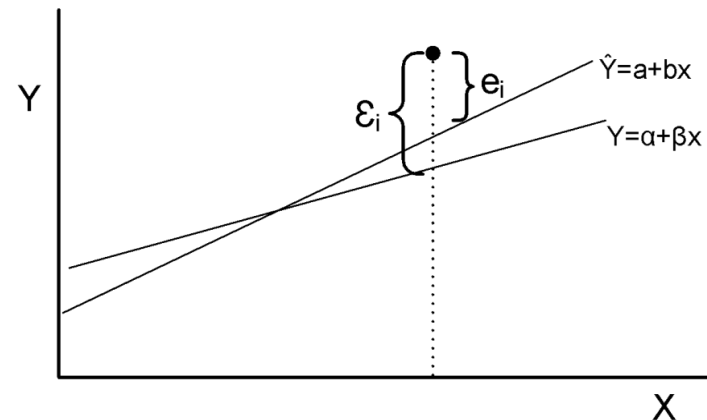
Least Square Method to estimate α and β

Sum of Squares Error (SSE)

The residual sum of squares is often called **the sum of squares of the errors** about the fitted line and is denoted as

SSE

$$\begin{aligned} SSE &= \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - a - bx_i)^2 \end{aligned}$$



We need to **minimize the value of SSE** and hence to determine the parameters of a and b .

Least Square Method to estimate α and β

Minimizing the Sum of Squares Error (SSE)

Step 1: Differentiation

Differentiating SSE with respect to a and b , we have

$$\frac{\partial(SSE)}{\partial a} = -2 \sum_{i=1}^n (y_i - a - bx_i)$$

$$\frac{\partial(SSE)}{\partial b} = -2 \sum_{i=1}^n (y_i - a - bx_i) \cdot x_i$$

Step 2: Equating the partial derivatives to zero

For minimum value of SSE, $\frac{\partial(SSE)}{\partial a} = 0$, and $\frac{\partial(SSE)}{\partial b} = 0$

Least Square Method to estimate α and β



Minimizing the Sum of Squares Error (SSE)

Step 2: Equating the partial derivatives to zero

For minimum value of SSE, $\frac{\partial(SSE)}{\partial a} = 0$, and $\frac{\partial(SSE)}{\partial b} = 0$

Thus we get,

$$na + b \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

Least Square Method to estimate α and β



Minimizing the Sum of Squares Error (SSE)

Step 2: Equating the partial derivatives to zero

For minimum value of SSE, $\frac{\partial(SSE)}{\partial a} = 0$,
and $\frac{\partial(SSE)}{\partial b} = 0$

Thus we get,

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$$a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

Step 3: Solving for a and b

These two equations on the left can be solved to determine the values of a and b , and it can be calculated that

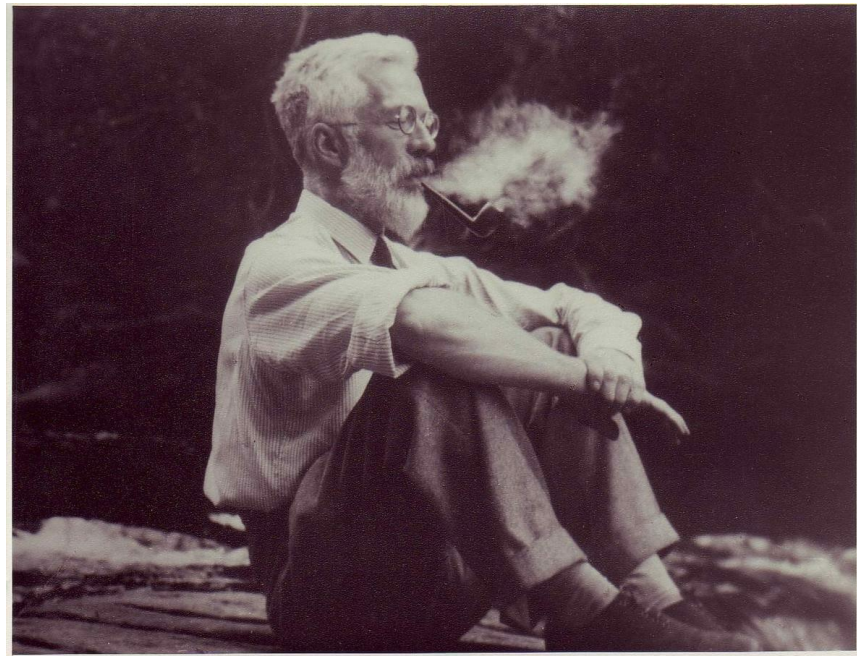
$$b = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$a = \bar{y} - b\bar{x}$$

Simple Linear Regression: Example

Heart Disease and Cigarettes

- Data on coronary heart disease and cigarette smoking in 21 developed countries (Landwehr and Watkins, 1987)
- Data have been rounded for computational convenience.



Do you know who is he?

Data

Surprisingly, the U.S. is the first country on the list-the country with the highest consumption and highest mortality.

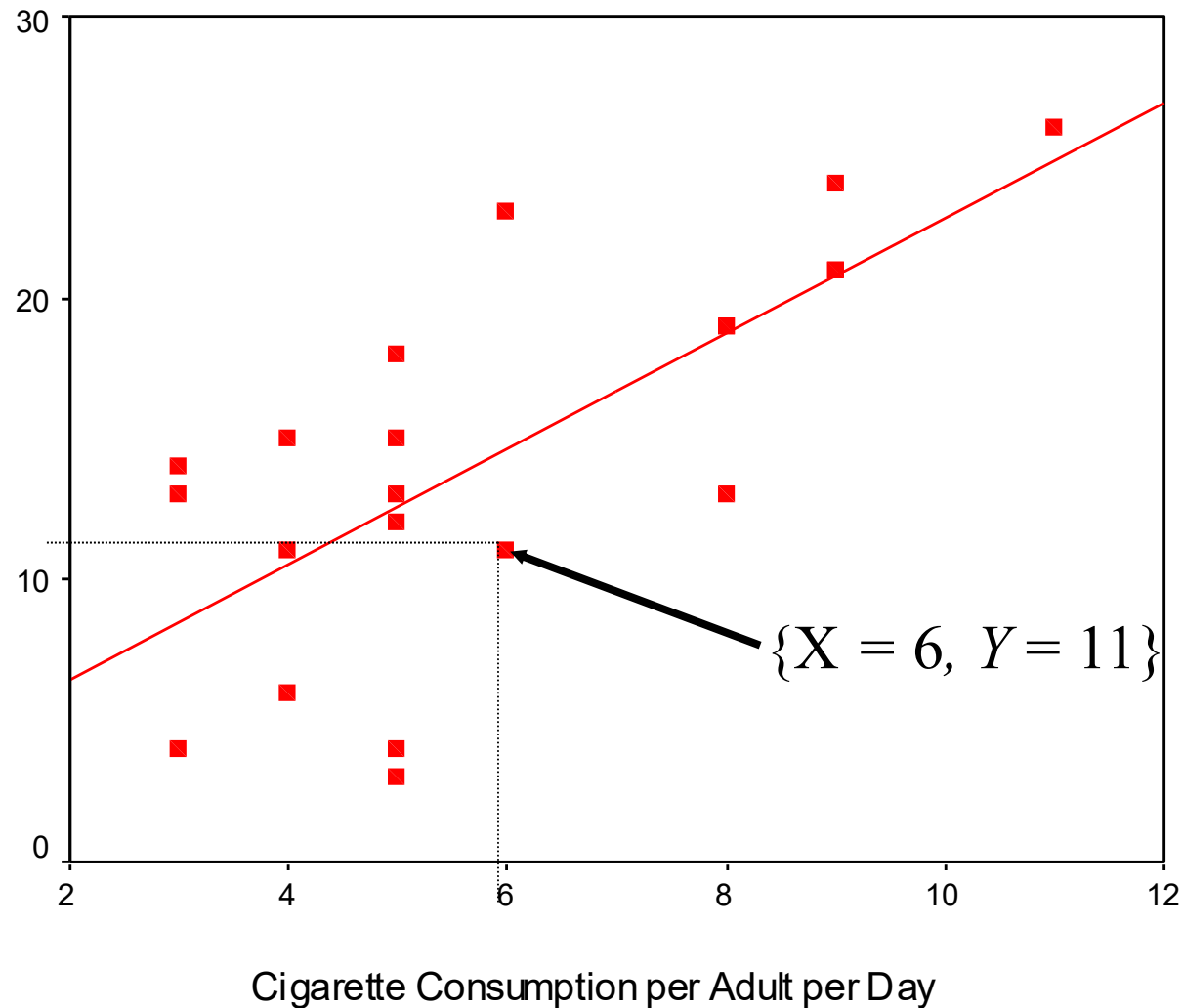
Scatterplot of Heart Disease

- Chronic Heart Disease (CHD) Mortality goes on ordinate (Y axis)
- Cigarette consumption on abscissa (X axis)
- Best fitting line included for clarity

Country	Cigarettes	CHD
1	11	26
2	9	21
3	9	24
4	9	21
5	8	19
6	8	13
7	8	19
8	6	11
9	6	23
10	5	15
11	5	13
12	5	4
13	5	18
14	5	12
15	5	3
16	4	11
17	4	15
18	4	6
19	3	13
20	3	4
21	3	14

What Does the Scatterplot Show?

- As smoking increases, so does coronary heart disease mortality.
- Relationship looks strong
- Not all data points on line.
- This gives us residuals” or “errors of prediction”



Important Formulae

- Correlation coefficient between two R.V.s X and Y , usually denoted by $r(X, Y)$ or r_{XY} is a numerical measure of linear relationship between them and is defined as:

$$r_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

- Remember that variance is:

$$Var_X = \frac{\sum (X - \bar{X})^2}{N - 1} = \frac{\sum (X - \bar{X})(X - \bar{X})}{N - 1}$$

- The formula for covariance is:

$$Cov_{XY} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{N - 1}$$

Simple Linear Regression Formula:

$$\hat{Y} = a + bX$$

$$b = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{Cov(X, Y)}{\sigma_X^2}$$

$$a = \bar{y} - b\bar{x}$$

Example: Heart Disease and Cigarettes

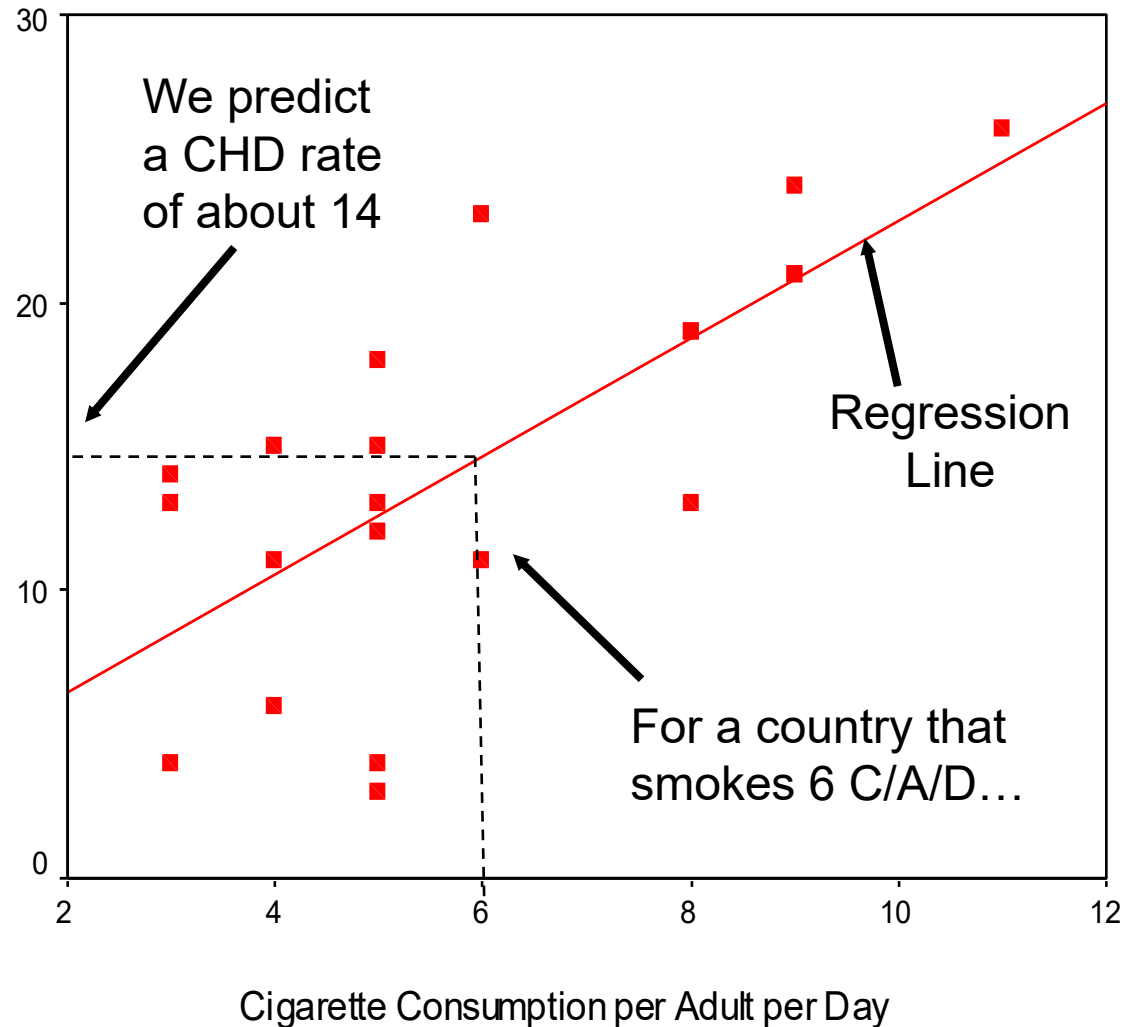
Country	X (Cig.)	Y (CHD)	$(X - \bar{X})$	$(Y - \bar{Y})$	$(X - \bar{X}) * (Y - \bar{Y})$
1	11	26	5.05	11.48	57.97
2	9	21	3.05	6.48	19.76
3	9	24	3.05	9.48	28.91
4	9	21	3.05	6.48	19.76
5	8	19	2.05	4.48	9.18
6	8	13	2.05	-1.52	-3.12
7	8	19	2.05	4.48	9.18
8	6	11	0.05	-3.52	-0.18
9	6	23	0.05	8.48	0.42
10	5	15	-0.95	0.48	-0.46
11	5	13	-0.95	-1.52	1.44
12	5	4	-0.95	-10.52	9.99
13	5	18	-0.95	3.48	-3.31
14	5	12	-0.95	-2.52	2.39
15	5	3	-0.95	-11.52	10.94
16	4	11	-1.95	-3.52	6.86
17	4	15	-1.95	0.48	-0.94
18	4	6	-1.95	-8.52	16.61
19	3	13	-2.95	-1.52	4.48
20	3	4	-2.95	-10.52	31.03
21	3	14	-2.95	-0.52	1.53
Mean	5.95	14.52			
SD	2.33	6.69			
Sum					222.44

$$Cov_{cig.&CHD} = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{N - 1} = \frac{222.44}{21 - 1} = 11.12$$

$$r = \frac{cov_{XY}}{s_X s_Y} = \frac{11.12}{(2.33)(6.69)} = \frac{11.12}{15.59} = .713$$

Country	Cigarettes	CHD
1	11	26
2	9	21
3	9	24
4	9	21
5	8	19
6	8	13
7	8	19
8	6	11
9	6	23
10	5	15
11	5	13
12	5	4
13	5	18
14	5	12
15	5	3
16	4	11
17	4	15
18	4	6
19	3	13
20	3	4
21	3	14

Example: Heart Disease and Cigarettes



Regression line

Formula:

$$\hat{Y} = a + bX$$

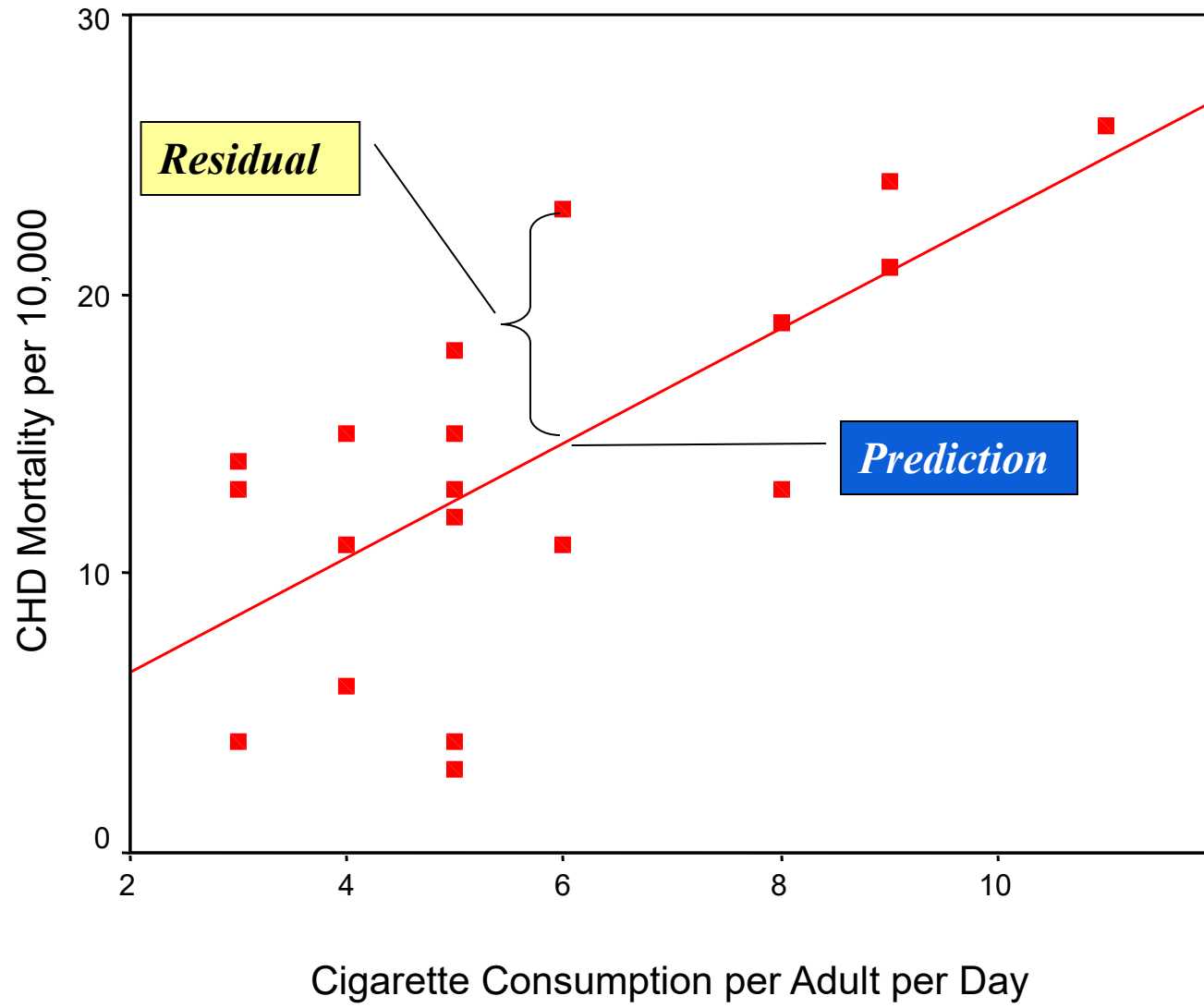
- $\hat{Y}$ = the predicted value of Y (e.g. CHD mortality)
- X = the predictor variable (e.g. average cig./adult/country)
- a and b are “Coefficients”
- b = slope ,i.e. change in predicted Y for one unit change in X
- a = intercept , i.e. value of $\hat{Y}$ when $X = 0$
- $Cov(X, Y) = 11.12619$
- $\sigma^2_X = 2.334014^2 = 5.447619$
- $b = 11.12619 / 5.447619 = 2.042395$
- $a = 14.52381 - 2.042395 * 5.952381 = 2.366696$
- the equation of the least-squares line:

$$\hat{y} = 2.367 + 2.042 x$$



Assumptions Revisited

- **Residuals need to be:**
 - Bell-shaped (normal)
 - Stable (over time)*
 - Random
 - Unrelated (we think X is related to Y) Plot your data before doing regression analysis
- Residuals need to show certain properties for regression to work properly
- The regression equation can be used to predict (or possibly manage) output data from input/process data
- Each residual is the difference between an observed y value and the corresponding predicted y value.
- **Plotting the residual:** A **residual plot** is a scatterplot of the $(x, \text{residual})$ pairs. Isolated points or a pattern of points in the residual plot indicate potential problems.
- Residuals are highly useful for studying whether a given regression model is appropriate for the data at hand.

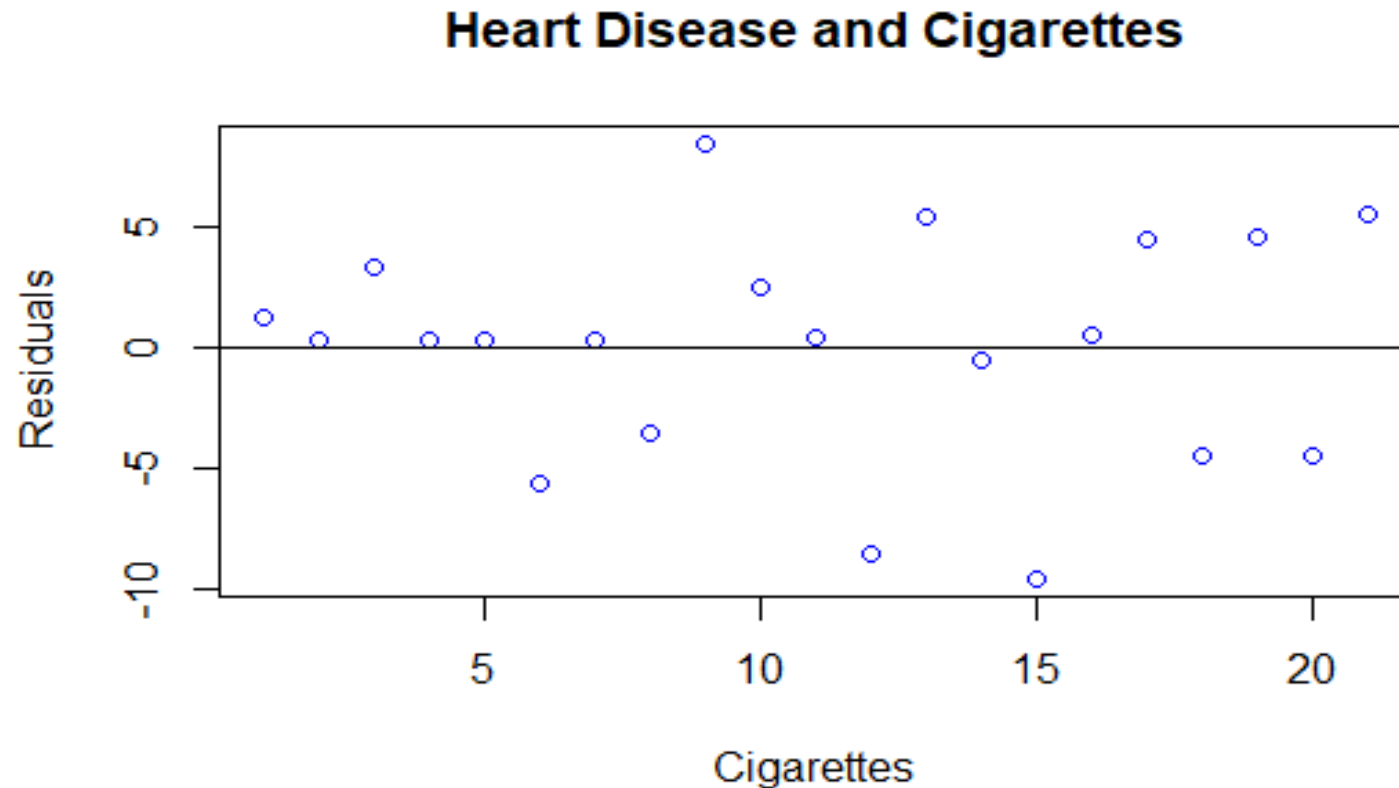


Residual : Heart Disease and Cigarettes

The equation of the least-squares line: $\hat{y} = 2.367 + 2.042 x$

	x	y	$\hat{y}$	$y - \hat{y}$
1	11	26	24.829	1.1669580
2	9	21	20.745	0.2517483
3	9	24	20.745	3.2517483
⋮	⋮	⋮	⋮	⋮
7	8	19	18.703	0.2941434
8	6	11	14.619	-3.6210664
9	6	23	14.619	8.3789336
⋮	⋮	⋮	⋮	⋮
13	5	18	12.577	5.4213287
14	5	12	12.577	-0.5786713
15	5	3	12.577	-9.5786713
⋮	⋮	⋮	⋮	⋮
18	4	6	10.535	-4.5362762
19	3	13	8.493	4.5061189
20	3	4	8.493	-4.4938811
21	3	14	8.493	5.5061189

Residual Plot: Heart Disease and Cigarettes



Residuals plots must be checked to ensure the assumptions hold; otherwise, the regression equation may be incorrect or misleading.

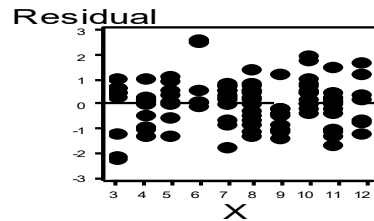
Checking Assumptions About Residuals

Residuals Plot

Good

1. Residuals vs Each X

Used to check that the residuals are not related to the Xs



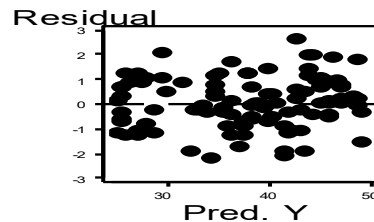
2. Time Plot of Residuals

Used to check for stability over time



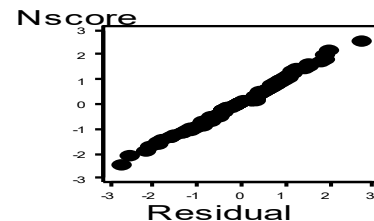
3. Residuals vs Predicted Y (Fits)

Used to check that they are constant over the range of Ys



4. Normal Probability Plot of Residuals

Used to check that residuals are Normal



Bad

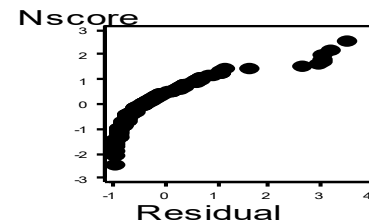
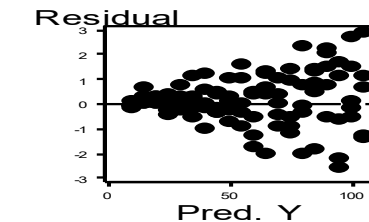
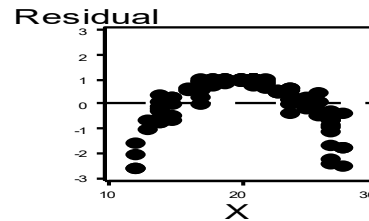
Meaning / Actions

The relationship between X & Y is not a straight line, but a curve. Try a transformation on X, Y, or both. Or use X^2 in a multiple regression.

Any pattern visible over time means another factor, related to time, influences Y. Try to discover it and include it in a multiple regression.

This fan shape means the variation increases as Y gets larger (it's not constant). Try a square root, log, or inverse transformation on Y.

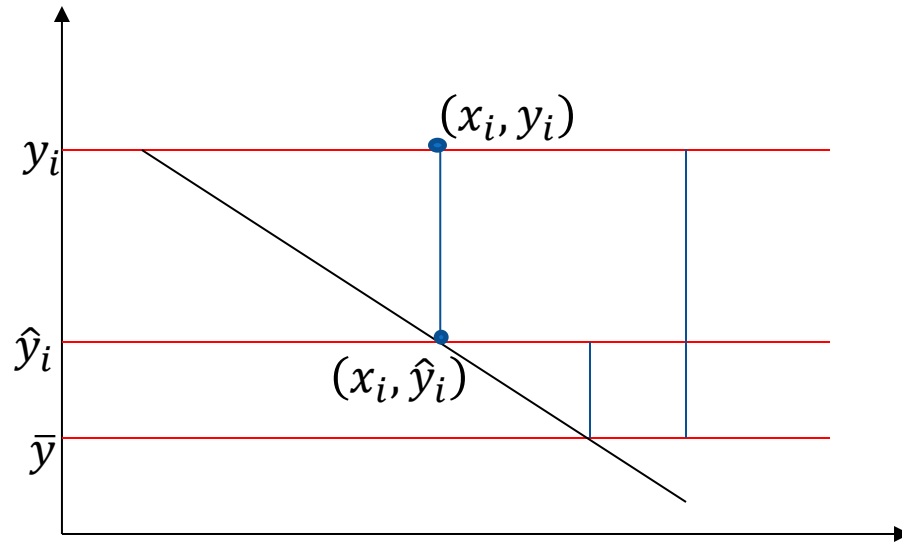
The residuals are not Normal. Try a transformation on X or Y or both.



**R^2 : Measure of Quality of
the Fitting**

Variation

- A set of data : (x_i, y_i) .
- $\hat{Y}$: predicted value of Y
- $\bar{Y}$: mean of Y – value



Deviation of the i^{th} observation from the mean = Deviation of the i^{th} observation from the predicted value + Deviation of the i^{th} predicted value from the mean

$$\Rightarrow (Y_i - \bar{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y})$$

- Total variation in data = $\sum_{i=1}^n (Y_i - \bar{Y})^2$.
- The sum of the squared of the differences between each predicted y-value and $\bar{Y}$ = $\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$
- The sum of the squared of the differences between the y-value of each ordered pair and each corresponding predicted y-value = $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$.

R^2 : Measure of Quality of Fit

- A quantity R^2 , is called **coefficient of determination** is used to measure the proportion of variability of the fitted model.
- We have $SSE = \sum_{i=1}^n (y_i - \hat{y})^2$
- It signifies the **variability due to error**.
- Now, let us define the **total corrected sum of squares**, defined as

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- SST represents the variation in the response values. The R^2 is

$$R^2 = 1 - \frac{SSE}{SST}$$

Note:

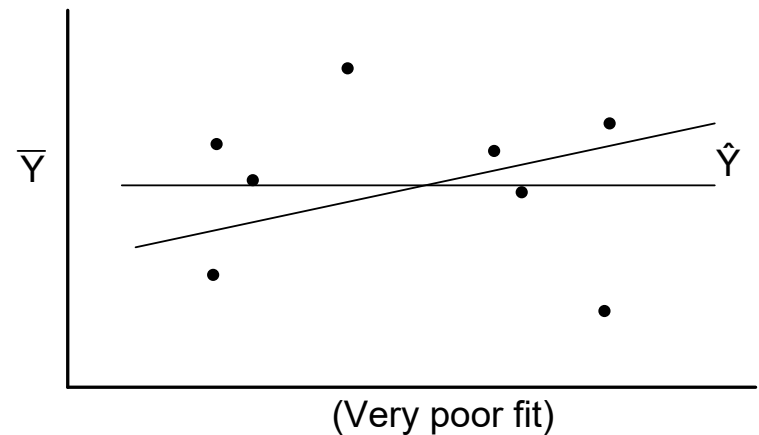
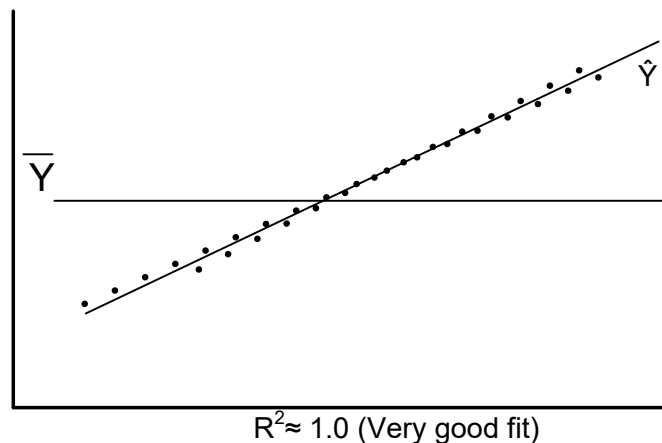
- If fit is perfect, all residuals are zero and thus $R^2 = 1.0$ (very good fit)
- If SSE is only slightly smaller than SST, then $R^2 \approx 0$ (very poor fit)
- The value of R^2 is often converted to a percentage (by multiplying by 100) and interpreted as the percentage of variation in y that can be explained by an approximate linear relationship between x and y .

R²: Measure of Quality Fit

Coefficient of Determination

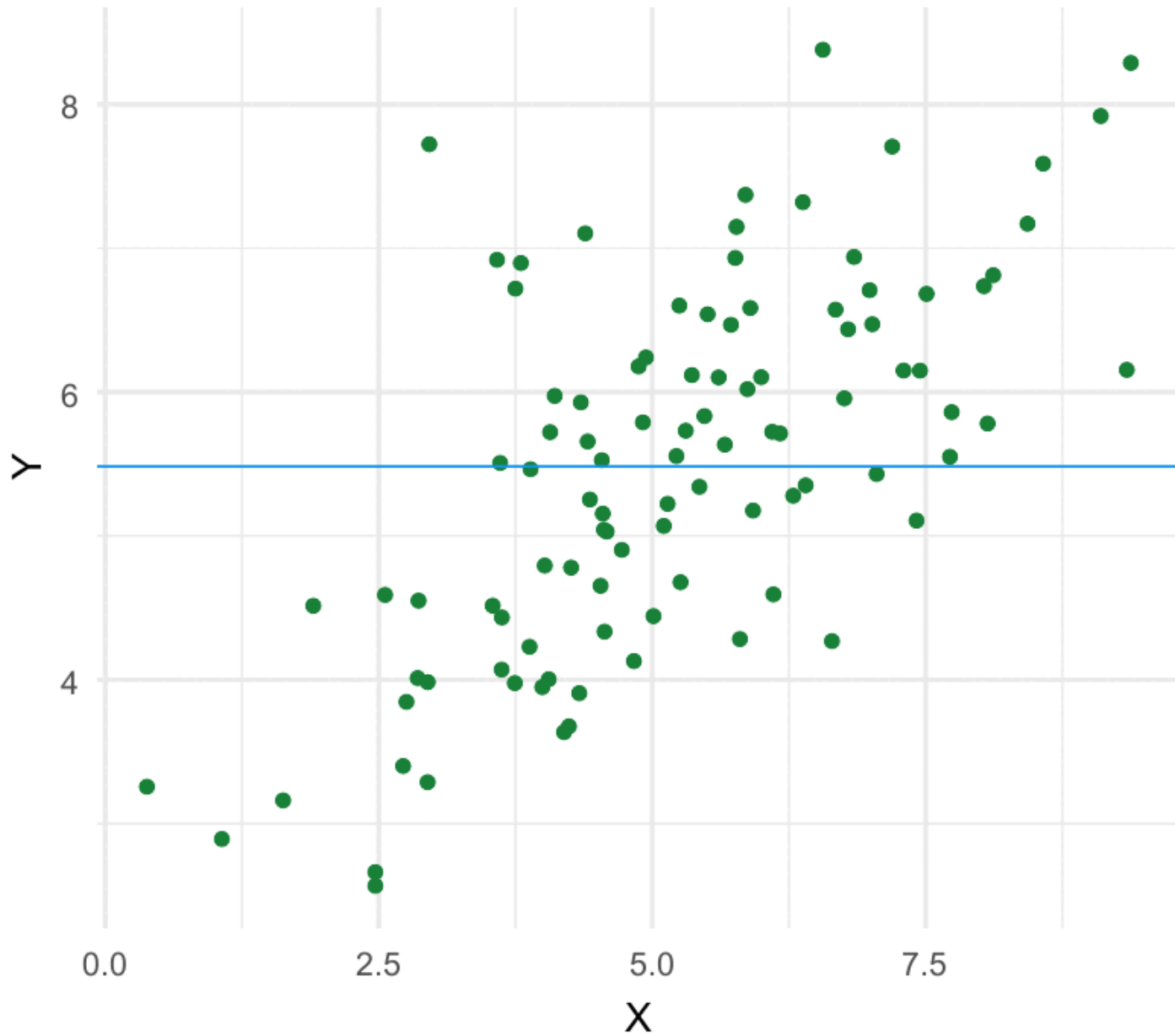
Note

- 🌟 If fit is perfect, all residuals are zero and thus $R^2 = 1.0$ (very good fit)
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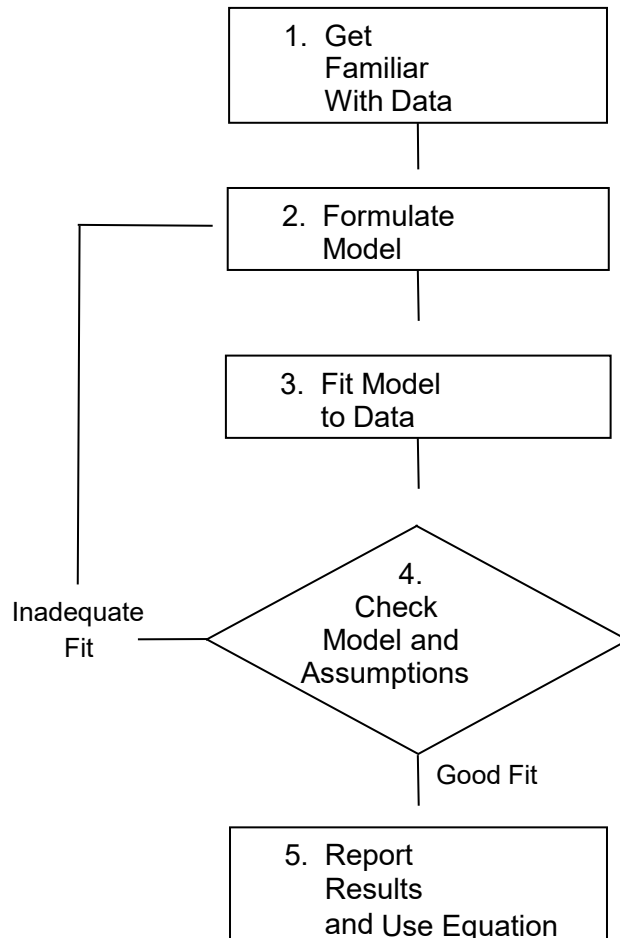


R-squared: 0

Angle: 0 degrees



Five Step Regression Procedure: Overview



- Look at plots
- Look at descriptive statistics

- Linear or curvilinear?
- One X or more Xs?
- Transform?
- Discrete X, discrete Y?

- Do the regression

- Look at residuals plots
- Look at unusual observations
- Look at R-Sq
- Look at P-values for b

- Make predictions for X-values of interest

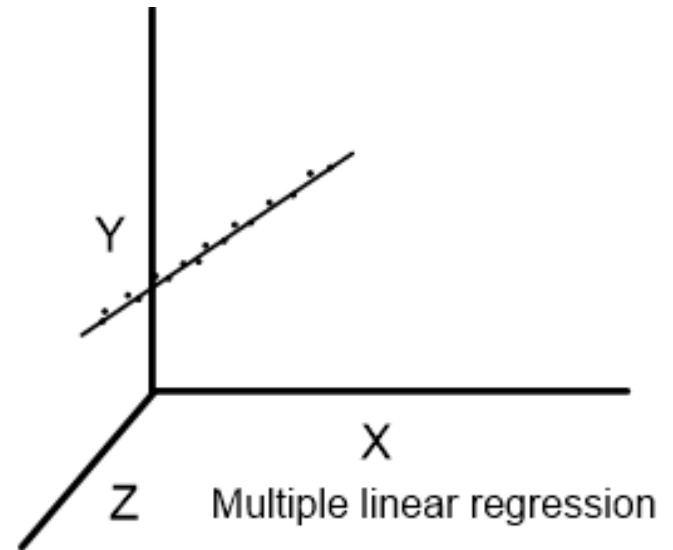


Multiple Linear Regression

Multiple Linear Regression

Definition:

- **Multiple Regression Model:** When more than one variable are independent variable, then the regression can be estimated as a multiple regression model
- **Multiple Linear Regression:** When this model is linear in coefficients, it is called multiple linear regression model



Multiple Linear Regression

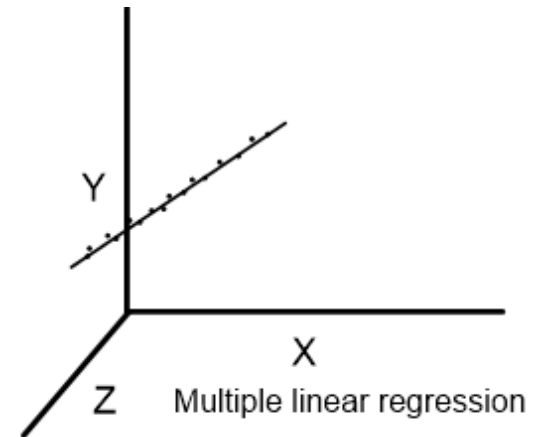
Formulation:

If k -independent variables $x_1, x_2, x_3, \dots, x_k$ are associated, the multiple linear regression model is given by

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_k x_k + \epsilon$$

And the estimated response is obtained as

$$\hat{y}_i = b_0 + b_1 x_1 + b_2 x_2 + b_k x_k$$





Multiple Linear Regression

Estimating the coefficients: The data points

Let the data points given to us are

$$(x_{1i}, x_{2i}, x_{3i}, \dots, x_{ki}, y_i) \quad i = 1, 2, \dots, n, \quad n > k$$

where y_i is the observed response to the values $x_{1i}, x_{2i}, x_{3i}, \dots, x_{ki}$ of k independent variables $x_1, x_2, x_3, \dots, x_k$.

Multiple Linear Regression

Estimating the coefficients: The model formulation

So, the regression model in this case is given by,

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_k x_{ki} + \epsilon_i$$

and $\hat{y}_i = b_0 + b_1 x_{1i} + b_2 x_{2i} + b_k x_{ki} + e_i$

where ϵ_i and e_i are the random error and residual error, respectively associated with true response y_i and fitted response $\hat{y}_i$.



Multiple Linear Regression

Estimating the coefficients: Minimization of the SSE

Using the concept of **Least Square Method** to estimate $b_0, b_1, b_2, \dots, b_k$, we minimize the expression

$$\text{SSE} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

To minimize the **SSE**, we need to differentiate **SSE** in turn with respect to $b_0, b_1, b_2, \dots, b_k$ and equate to zero.

Multiple Linear Regression

Estimating the coefficients: Minimization of the SSE

Differentiating SSE in turn with respect to $b_0, b_1, b_2, \dots, b_k$ and equating to zero, we generate the set of $(k+1)$ normal **estimation equations for multiple linear regression**.

$$nb_0 + b_1 \sum_{i=1}^n x_{1i} + b_2 \sum_{i=1}^n x_{2i} + \dots + b_k \sum_{i=1}^n x_{ki} = \sum_{i=1}^n y_i$$

$$b_0 \sum_{i=1}^n x_{1i} + b_1 \sum_{i=1}^n x_{1i}^2 + b_2 \sum_{i=1}^n x_{1i} \cdot x_{2i} + \dots + b_k \sum_{i=1}^n x_{1i} \cdot x_{ki} = \sum_{i=1}^n x_{1i} \cdot y_i$$

$$\begin{array}{cccccc} \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

$$b_0 \sum_{i=1}^n x_{ki} + b_1 \sum_{i=1}^n x_{ki} \cdot x_{1i} + b_2 \sum_{i=1}^n x_{ki} \cdot x_{2i} + \dots + b_k \sum_{i=1}^n x_{ki}^2 = \sum_{i=1}^n x_{ki} \cdot y_i$$

The system of linear equations can be solved for $b_0, b_1, \dots, b_k$ by any appropriate method for solving system of linear equations.

Mathematical Formulation of the model

A multiple linear regression model takes the form

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_{k-1} x_{k-1} + \varepsilon \quad \dots \dots \dots (1)$$

- Y is the response variable
- $x_1, x_2, \dots, x_{k-1}$ are $(k - 1)$ explanatory variables
- $\beta_0, \beta_1, \beta_2, \dots, \beta_{k-1}$ are regression coefficients.
- The error ε is assumed to be *iid* with mean 0 and variance σ^2 .
- For Hypothesis testing and the setting of confidence limits, we also assume that ε is normally distributed.
- The parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_{k-1}$ are sometimes called partial regression coefficients because β_j , represents the mean change in Y per unit change in x_j while all the other x variables are held constant.
- The linearity of the model (1) is defined with respect to the regression coefficients.
- We assume that the explanatory variables are known (controlled variable) and the error-free but the response variable is treated as random variable.

Linear Least Square Solution using Matrix Method

- The first objective is to estimate the k unknown parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_{k-1}$.
- Suppose that we have n sets of observations of all the variables $(y_i, x_{i1}, x_{i2}, \dots, x_{ik-1}), i = 1, 2, \dots, n$.
- The multiple regression model can be written as

- $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

- Or, equivalently,

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k-1} \\ 1 & x_{21} & x_{22} & \dots & x_{2k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk-1} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{k-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

- The vector of mean values $E[\mathbf{Y}]$ of $\mathbf{Y}$ is

$$E[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}. \text{ (since } E(\varepsilon) = 0 \text{)}$$

Linear Least Square Solution using Matrix Method

- The least squares solution is obtained by minimizing

$$\begin{aligned} SSResid &= \sum_{i=0}^n e_i^2 = \mathbf{e}^T \mathbf{e} = (\mathbf{Y} - \hat{\mathbf{Y}})^T (\mathbf{Y} - \hat{\mathbf{Y}}) \\ &= (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \end{aligned}$$

- By minimizing the above equation, we can find

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y},$$

- $\mathbf{Y}$ is a $(n \times 1)$ vector of observed values.
- If the $(k \times k)$ matrix $\mathbf{X}^T \mathbf{X}$, which is symmetric, can be inverted the least squares solution to the unknown parameters is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y},$$

which is a $(k \times 1)$ vector of fitted parameters.

- The residual $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, which constitute a $(n \times 1)$ vector of differences between the observed and estimated mean values of Y are taken as estimators of the errors

$$\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$$

and are used in the assessment of the model.

Linear Least Square Solution using Matrix Method

Estimation of Model parameters:

LSM determines the parameters by minimizing

(Least square method) $SS_{Res} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \dots + \hat{\beta}_{K-1} X_{iK-1}$

$$SS_{Res} = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \hat{\beta}_2 X_{i2} - \dots - \hat{\beta}_{K-1} X_{iK-1})^2$$

AH. Method:

In matrix form:

$e = (Y - \hat{Y})$; where $e = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}$, $X, \hat{Y}$.

$$SS_{Res} = \sum_{i=1}^n e_i^2 = e'e = (Y - \hat{Y})'(Y - \hat{Y})$$

$$= (Y - X\hat{\beta})'(Y - X\hat{\beta}) = Y'Y - Y'X\hat{\beta} - \hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$$

[since $\hat{Y} = X\hat{\beta}$]

(1x1) scalar quantity and equal

$$SS_{Res} = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$$

Normal equations:

$$\frac{\partial SS_{Res}}{\partial \hat{\beta}_0} = 0 \Rightarrow \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \dots - \hat{\beta}_{K-1} X_{iK-1}) = 0$$

$$\Rightarrow \sum_{i=1}^n e_i = 0$$

$$\frac{\partial SS_{Res}}{\partial \hat{\beta}_1} = 0 \Rightarrow \sum e_i X_{i1} = 0$$

$$\vdots$$

$$\frac{\partial SS_{Res}}{\partial \hat{\beta}_{K-1}} = 0 \Rightarrow \sum e_i X_{iK-1} = 0$$

K constraints
K normal equations (independent)

AH. Method: (In Matrix form):

$$\frac{\partial SS_{Res}}{\partial \hat{\beta}} = 0$$

$$\Rightarrow -2X'Y + 2X'X\hat{\beta} = 0$$

$\Rightarrow$

$$\Rightarrow \hat{\beta} = (X'X)^{-1} (X'Y) = \text{Least square estimation of } K \text{ unknown parameters for multiple linear regression model.}$$

Solving these K normal equations, we can have K unknown parameters.

Tackling Multicollinearity: Ridge regression

- When there are near-linear dependence among the regressors, the problem of multicollinearity is said to exist (the columns of the matrix $X^T X$ are linearly dependent).

In least square regression, the coefficients β 's of x variables are identified by minimizing the residual sum of squares

$$RSS = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2$$

In ridge regression, the coefficients β 's of x variables are identified by minimizing a slightly different quantity

$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 = RSS + \lambda \sum_{j=1}^p \beta_j^2$$

Where $\lambda \geq 0$ is a turning parameter and $\lambda \sum_{j=1}^p \beta_j^2$ is the shrinkage penalty,

which will be small when $\beta_1, \beta_2, \dots, \beta_p$ are close to zero.

References

